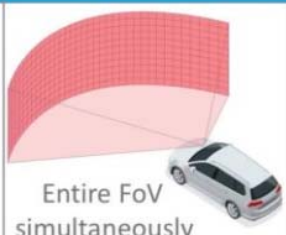
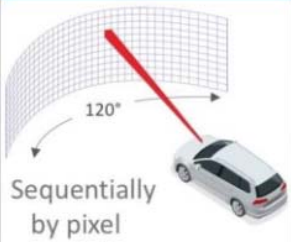
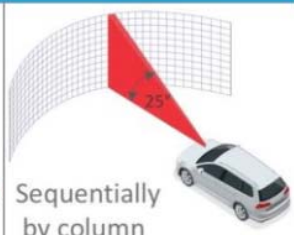
Computer Vision
Lecture 9: LiDAR Imaging

Yeungnam University
MSP lab.
Ho-Youl JUNG

영남대학교 Yeungnam University

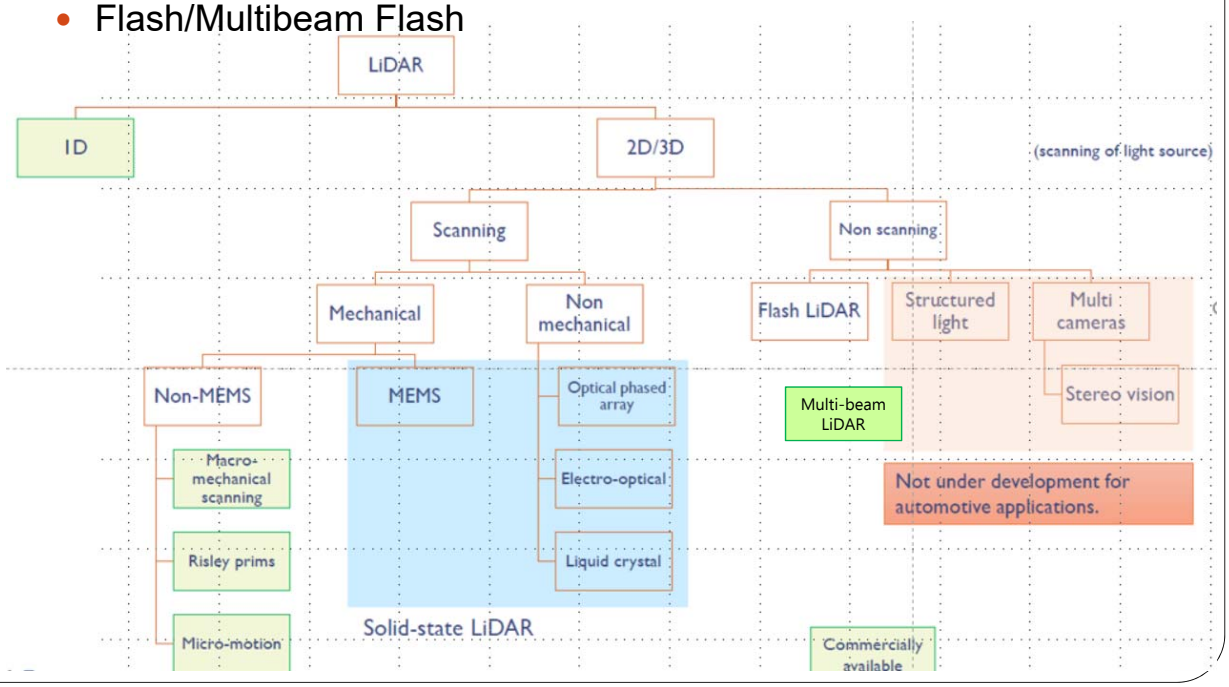
Lidar Imaging Strategies

- Flash
- 2D Raster-Scan
- 1D Line-Scan

	Flash	2D Raster-Scan	1D Line-Scan
Illumination pattern	 Entire FoV simultaneously	 Sequentially by pixel	 Sequentially by column
Returned signal per pixel	Low	Highest	High
Range	Too short (< 50 m)	Longest (> 250 m)	Long (250 m)
Frame rate	Fastest (> 30 fps)	Too slow (2 fps*)	Fast (30 fps)
Suitability	✗	✗	✓

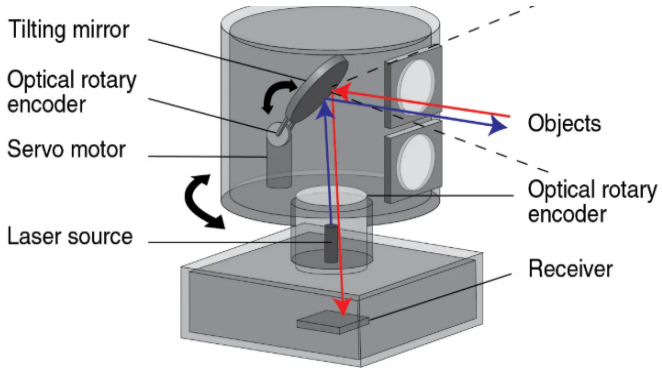
Lidar Imaging Strategies

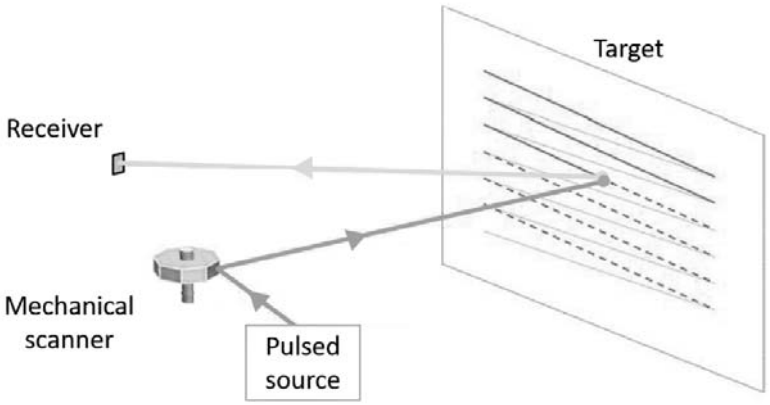
- Scanning
  - Mechanical Scanning
  - MEMS Scanning
  - OPAs
- Flash/Multibeam Flash



Mechanical Scanning

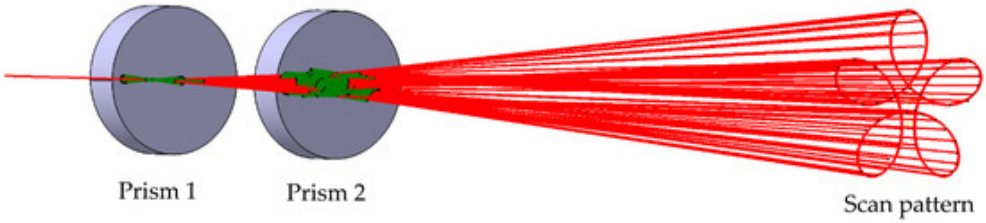
- Rotate 360 degree
- High power consumption (Motor)
- Limited frame rates(1~100Hz)
- Questionable in reliability and maintenance
- Not rigid to shock and vibration





Risley Prism Scanning

- Two rotating, wedge-shaped prisms
- Compact and Lightweight: mechanically simple, relying on small rotating prisms rather than bulky mirrors.
- High Durability: less susceptible to vibration and shock
- Flexible Scanning Patterns
- Complex Control Algorithms
- Non-Uniform Point Cloud uniform 360-degree scan.
- Limited Field of View: a smaller field of view (FOV)

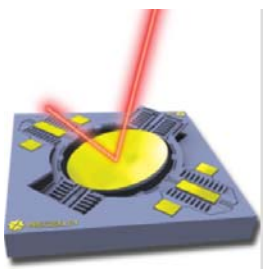


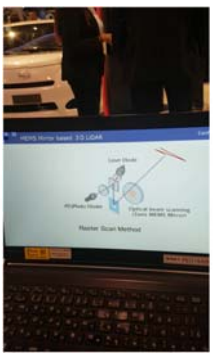
(c)

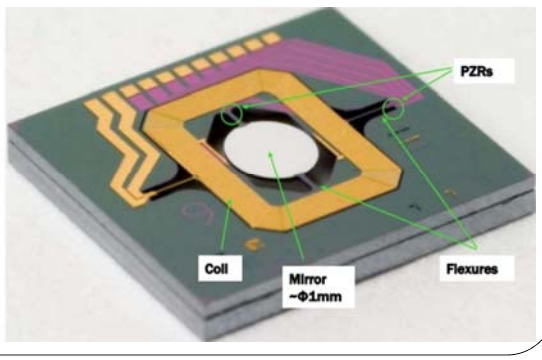
Duma, V., & Dimb, A. (2020). Exact Scan Patterns of Rotational Risley Prisms Obtained with a Graphical Method: Multi-Parameter Analysis and Design. *Applied Sciences*, *11*(18), 8451. https://doi.org/10.3390/app11188451

MEMS Scanning

- Micro electro mechanical systems (MEMS)-based Lidar scanners
- Programmable control of laser beam position using tiny mirrors with only a few mm in diameter
- Reduced size (lightweight, compact), low power consumption
- Two classes according to their operating mechanical mode: resonant and non-resonant
  - Non-resonant MEMS mirrors (also called quasi-static MEMS mirrors) : large degree of freedom in the trajectory design, rather complex control, limited scan angle
  - Resonant MEMS mirrors: large scan angle, relatively simple control design, sinusoidal scan trajectory(the scan speed is not uniform)
- Need careful laser power handling to avoid mirror damage

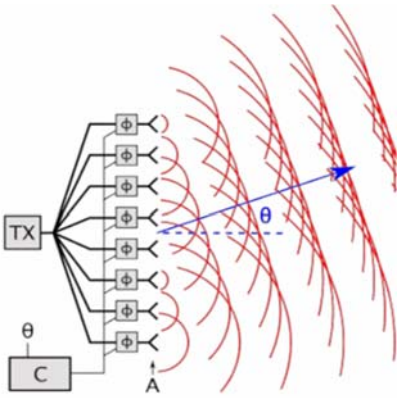


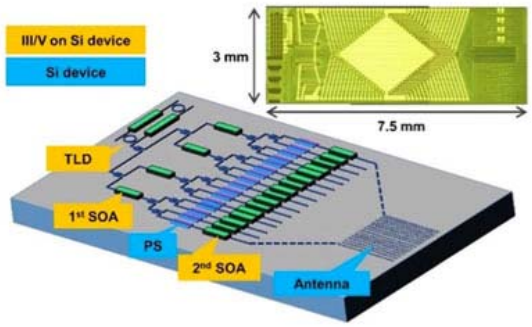




OPA Scanning

- Optical Phased Array (OPA)
- Solid state device: no mechanical moving parts
- Highly compact, single chip, large steering range at high speed
- Insertion loss of the laser power
- Need further development from its current state





https://www.drivingvisionnews.com/news/2022/07/06/samsungs-single-chip-opa-lidar-prototype/

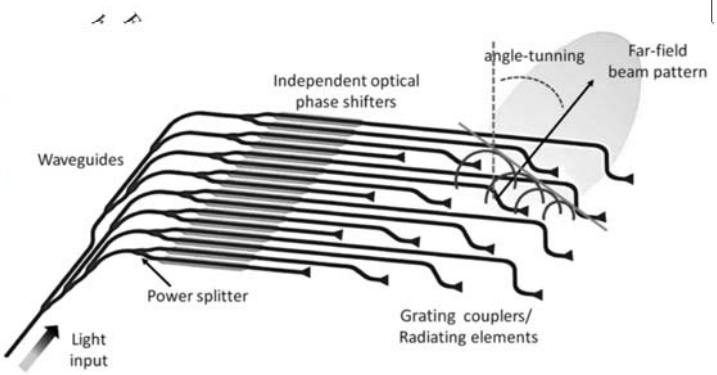


Fig. 2. Simplified diagram of an integrated silicon photonics OPA transmitter including an input optical signal which is routed on-chip using dielectric waveguides, a power splitter tree, phase shifter network, and an array of radiating elements.

Flash

- Transmitting optical elements illuminate a whole scene and a linear array (or matrix) of detectors receives the signals of separate angular subsections
- Emitted light pulse is dispersed in all directions, significantly reducing the SNR
  - Pulse based Flash
  - AMCW based Flash(ToF Camera)
  - FMCW based Flash

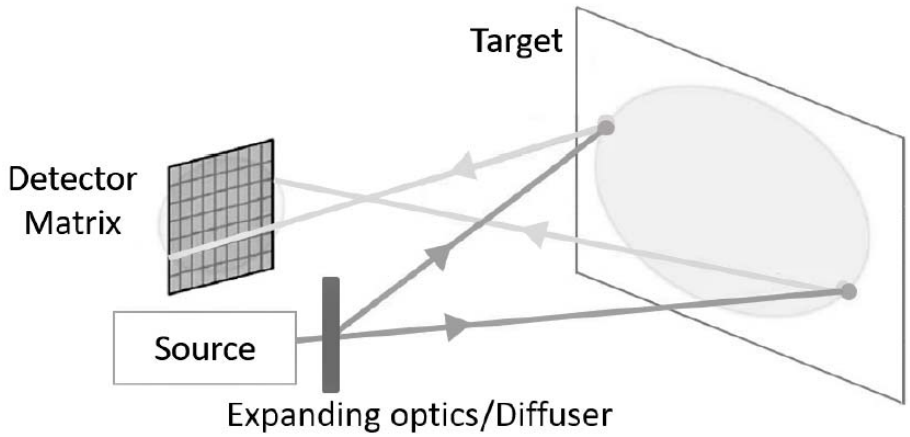
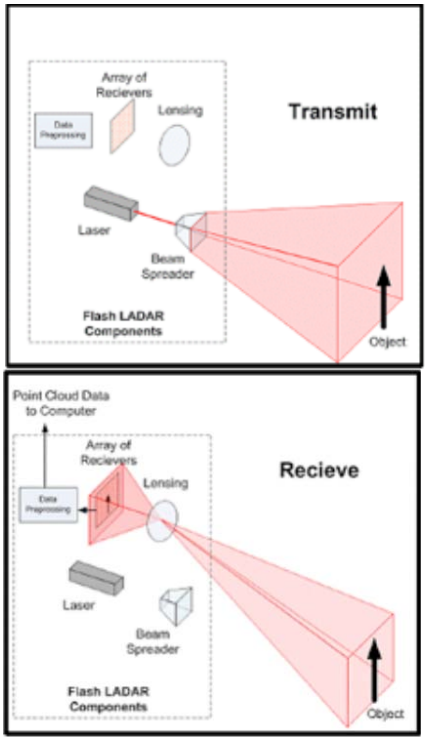


Figure 8. Detector array-based lidar diagram.

Pulse based Flash

- Using an appropriate divergent optical system which expands the laser beam to illuminate the full FOV
- Each detector in the array is individually triggered by the arrival of a pulse return, and measures both its intensity and the range
- Very high peak illumination power and very sensitive detectors are required in comparison to single-pixel scanners
- Limited to measurement distance(20 m to 150 m) for eye-safety considerations(emitted power)
- Detectors are usually SPADs (single photon avalanche diodes)
- Retro-reflectors flood the SPAD detector with photons, saturating it, and blinding the entire sensor for some frames



Comparison of Lidar Imaging Strategies

	Mechanical Scanners	MEMS Scanners	OPAs	Flash
Working principle	Galvos, rotating mirrors or prisms	MEMS micromirror	Phased array of antennas	Pulsed flood illumination
Main advantage	360 deg FOV in horizontal	Compact and lightweight	Full Solid State	Fast frame rate
Main disadvantage	Moving elements, bulky	Laser power management, linearity	Lab-only for long-range	Limited range/Blindable

Sources

- Fiber lasers

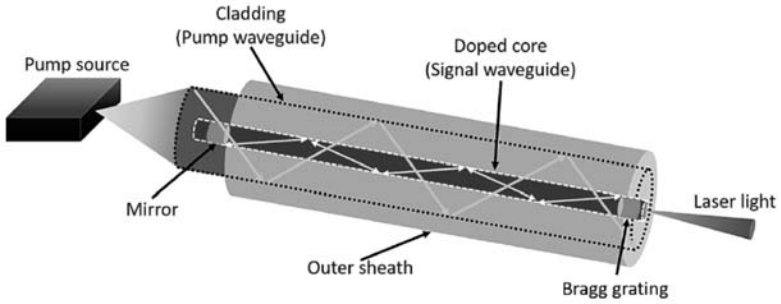


Figure 9. Schematic diagram of the fiber laser.

- Microchip lasers

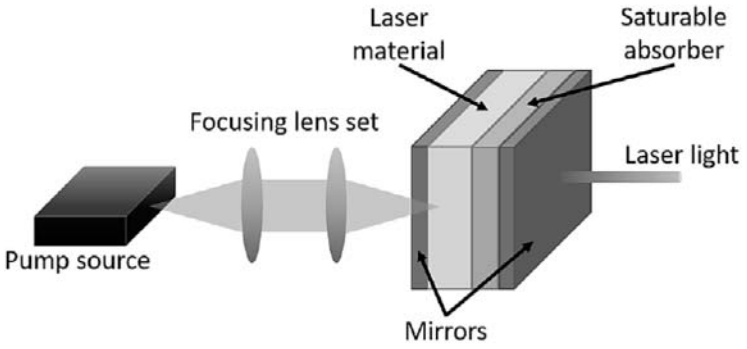
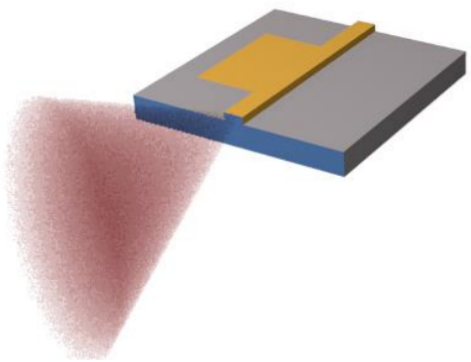


Figure 10. Schematic diagram of a microchip laser.

Sources

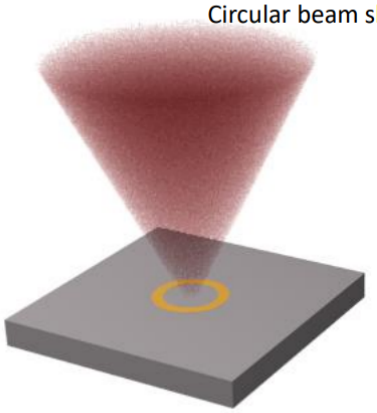
- **Diode lasers**
  - edge-emitting lasers (**EEL**), vertical-cavity surface-emitting lasers (**VCSEL**), distributed feedback lasers (DFB), high-power stacked diode bars or external cavity DL

Mirrors formed by facets,
Waveguide typically formed by ridge and wafer structure,
Facet coatings required,
Elliptical beam shape.



(Horizontal Cavity) Edge Emitting Laser - EEL

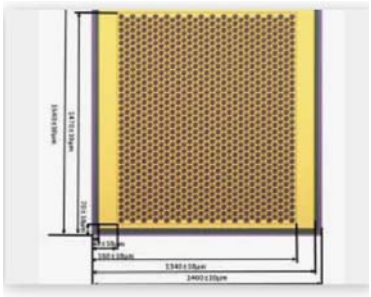
Mirrors grown in wafer structure,
Waveguide formed by oxidation,
Facet coatings not required,
Circular beam shape.



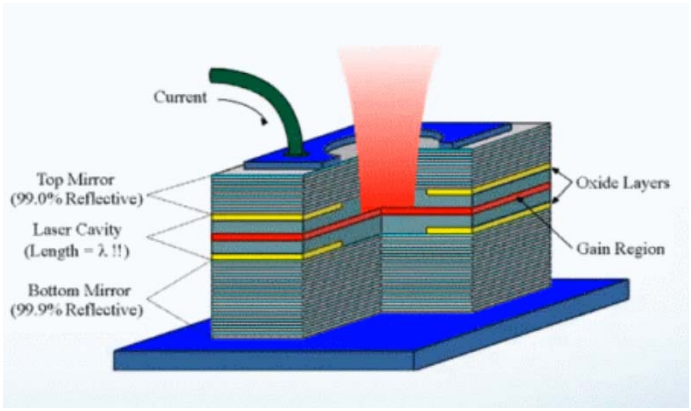
Vertical Cavity Surface Emitting Laser - VCSEL

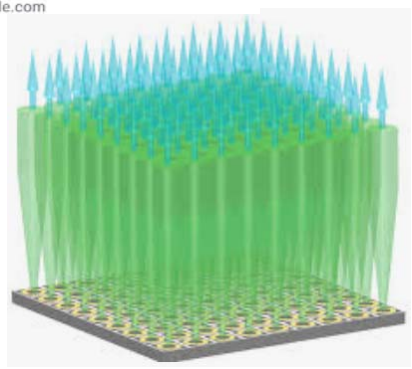
Sources

- **Diode lasers**
  - **VCSEL** (Vertical Cavity Surface Emitting Laser, 수직 공진 표면 발광 레이저) : a type of semiconductor laser diode
  - VCSELs for wavelengths from 650 nm to 1300 nm are typically based on gallium arsenide (GaAs) wafer
  - VCSEL array (>20,000)



Laser VCSEL Array 4W At 940nm Suppliers & ...
brandnewdiode.com





Comparisons

Table 3. Summary: main features of sources for lidar.

	Fibre Laser	Microchip Laser	Diode Laser
Amplifying media	Doped optical fibre	Semiconductor crystal	Semiconductor PN junction
Peak power (typ)	>10 kW	>1 kW	0.1 kW
PRR	<1 MHz	<1 MHz	≈100 KHz
Pulse width	<5 ns	<5 ns	100 ns
Main advantage	Pulse peak power, PRR, beam quality. Beam delivery	Pulse peak power, PRR, beam quality	Cost, compact
Main disadvantage	Cost	Cost, beam delivery	Max output power and PRR. Beam quality

Note: these are typical values for reference when used for lidar. Individual components evolve continuously and may be different at the time of publishing or from specific providers. Further, several trade-offs related to laser performance need be considered (e.g., some DLs may have much larger peak power, but at smaller PRR values).

pulse repetition rate (PRR)

Photodetectors

- Detection sensitivity of the wavelength is determined by the material of the detector
  - Si-based detectors are used for the wavelength 300 nm~1,100 nm
  - InGaAs detectors are used the wavelength above 1,100 nm
- Five different types of detectors
  - PIN diodes, avalanche photodiodes (APD), single-photon avalanche photodiodes (SPAD), multi-pixel photon counters (MPPC), photomultiplier tubes (PMT).

Comparisons

Table 4. Summary: main features of photodetectors for lidar.

	PIN	APDs	SPADs	MPPCs	PMTs
Solid state	Yes	Yes	Yes	Yes	No
Gain (typ)	1	Linear (≈200)	Geiger (10^4)	Geiger (10^6)	Avalanche (10^6)
Main advantage	Fast	Adjustable gain by bias	Single photon detection	Single photon counting	Gain, UV detection
Main disadvantage	Limited for low SNR	Limited gain	Recovery time	Saturable, bias voltage dependence	Bulky, low QE, high voltage, magnetic fields

Note: these are typical values for reference. Individual components evolve continuously and may be different at the time of publishing or from specific providers.

Ref

[1] Santiago Royo and Maria Ballesta-Garcia, “An Overview of Lidar Imaging Systems for Autonomous Vehicles”, Applied Sciences, September 2019

[2] H.W. Yoo, N. Druml, D. Brunner, C. Schwarzl, T. Thurner, M. Hennecke, G. Schitter OVE, “MEMS-based lidar for autonomous driving”, Elektrotechnik & Informationstechnik (2018) 135/6: 408–415.

[3] YUNG-PENG CHANG, CHUN-NIEN LIU, ZINGWAY PEI, SHU-MING LEE, YEONG-KANG LAI, PIN HAN, HSING-KUN SHIH, AND WOOD-HI CHENG, “New scheme of LiDAR-embedded smart laser headlight for autonomous vehicles”, Vol. 27, No. 20, September 2019, Optics Express

[4] Low Cost LiDAR System Design Considerations : https://www.youtube.com/watch?v=N7gnoxVrjR8

[5] SensL Solid State LiDAR design consideration : https://www.youtube.com/watch?v=npnAr1BlQhw

[6] The Next Generation of SPAD Arrays for Automotive LiDAR : https://www.youtube.com/watch?v=upBYO_KS_Is

[7] “LiDAR systems: costs, integration, and major manufacturers”, 05.03.2020 |Author /Editor:Seth Lambert/ Erika Granath, https://www.mes-insights.com/lidar-systems-costs-integration-and-major-manufacturers-a-908358/

[8] Dr Khasha Ghaffarzadeh, “Automotive Lidar Market: Battlefield of a Hundred Technology Suppliers”, Mar 16, 2020 https://www.idtechex.com/en/research-article/automotive-lidar-market-battlefield-of-a-hundred-technology-suppliers/20156

[9] Automotive Lidar 2020, http://www.automotivelidar.com/

Thank You

Next : Automotive LiDAR Sensor

Automotive LiDAR Players

- Velodyne (United States)
- Ouster (United States)
- Aeva (United States)
- Luminar (United States)
- Baraja (Australia)
- Innoviz (Isreal)
- Valeo (France)
- Sick (Germany)
- Infineon (Germany)
- Benewake (China)
- Livox (China)
- Denso (Japan)
- Hokuyo (Japan)
- Pioneer (Japan)
- Xenomatix (Belgium)
- Ibeo (Germany)
- Blickfeld (Germany)
- Continental (Germany)
- LeddarTech (Canada)
- Quanergy (United States)
- AEye (United States)
- Robosense (China)
- Hesai (China)